

preparing interpretive reports—have high success rates. These occupations have low task coverage because AI can’t do the hands-on or administrative work in their job profiles, but it succeeds on the core knowledge work that dominates their workday.

Microbiologists fall below the 45-degree line, suggesting lower effective AI coverage than would be predicted by task coverage alone. Claude covers half of their tasks, but not their most time-intensive: hands-on research using specialized lab equipment.

This measure arguably gives a more realistic picture of job-level AI penetration. However, its implications depend on how often these Claude conversations actually displace or augment work that would otherwise be done by humans. For data entry clerks, AI likely does substitute for tasks previously performed manually. But when a Claude conversation maps to a teacher performing a lecture, it is less clear how this translates to reduced lecture time on the job. In future work, we could leverage our 1P API data to understand which of these tasks are being integrated into production workflows.

AI’s impact on the task content of jobs

Beyond how much of a worker’s day AI can successfully perform, a separate question is which tasks get covered, and whether those tend to be the high-skill or low-skill components of the job. Recent research has studied changes in the task mix within jobs to understand AI’s impact on wages and employment ([Autor and Thompson 2025](#); [Hampole et al 2025](#)). A key insight is that automation’s effects depend not just on how many tasks are covered, but on which tasks.

To see how jobs change when we remove the tasks AI can perform, we first construct a measure of the level of skill required for each task. O*NET doesn’t provide task-level education requirements, so we train a model that predicts years of schooling from task embeddings, using the BLS’s occupation-level education as the target.⁴ This way, a low-education occupation may still have a high-skill task if it looks like those that tend to exist in high-education occupations. For example, Legal Secretaries is a 12-year education occupation, but the task “Review legal publications and perform database searches to

identify laws and court decisions relevant to pending cases” is predicted to require 17.7 years because it resembles tasks typically performed by lawyers and paralegals.

The data shows that Claude tends to cover tasks that require higher levels of education. The mean predicted education for tasks in the economy is 13.2 years. For tasks that we see in our data, the mean prediction is about a year higher, 14.4 years (corresponding to an Associate’s degree). This aligns with the occupation-level results from earlier reports, showing more Claude usage among white collar occupations.

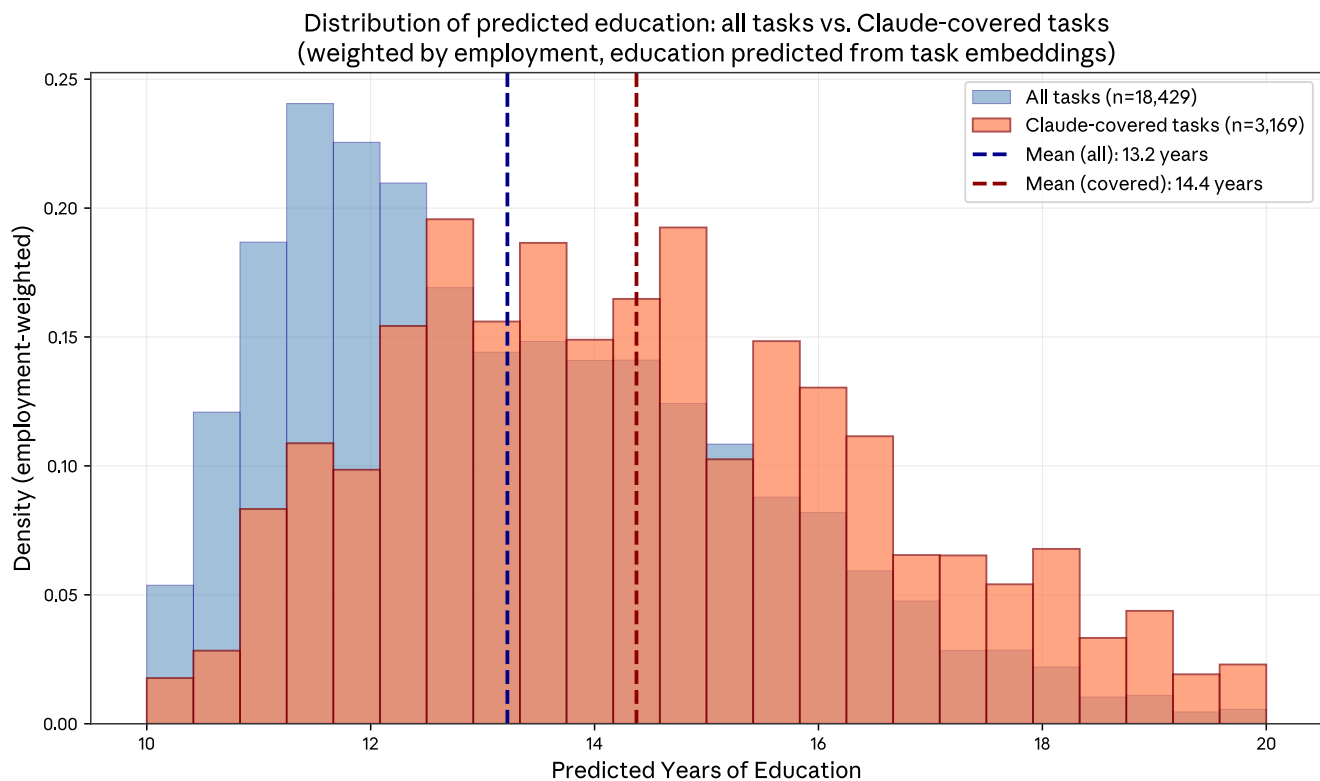


Figure 4.5: Education level of all tasks vs. Claude-covered tasks

This shows two histograms. The blue bars give the distribution of the predicted task-level education required for all tasks in the O*NET database, weighted by employment. The orange bars show the same, restricting to tasks that appear in Claude.ai data.

We next calculate how removing AI-covered tasks shifts the average education level of what remains. Overall, the net first-order impact is to deskill jobs, since AI removes tasks that require relatively higher levels of education. One job that experiences such deskilling is technical writers, which loses tasks like “Analyze developments in specific field to determine need for revisions” (18.7 years) and “Review published materials and recommend revisions or changes in scope, format” (16.4 years), leaving tasks like “Draw sketches to illustrate specified materials” (13.6 years) and “Observe production, developmental, and

experimental activities” (13.5 years). Travel agents also experience deskilling because AI covers tasks like “Plan, describe, arrange, and sell itinerary tour packages” (13.5 years) and “Compute cost of travel and accommodations” (13.4 years), while tasks like “Print or request transportation carrier tickets” (12.0 years) and “Collect payment for transportation and accommodations” (11.5 years) remain. Several teaching professions experience deskilling because AI addresses tasks like grading, advising students, writing grants, and conducting research without being able to do the hands-on work of delivering lectures in person and managing a classroom.

Some jobs see average education levels increase. Real estate managers experience upskilling because AI covers routine administrative tasks—maintaining sales records (12.8 years), reviewing rents against market rates (12.6 years)—while tasks requiring higher-level professional judgment and in-person interaction remain, like securing loans, negotiating with architecture firms, and meeting with boards.

These patterns illustrate how jobs may evolve over the coming years as their task content adjusts in response to AI. If the education level can be interpreted like expertise in [Autor and Thompson](#)’s analysis, their framework might predict that wages will fall and employment will increase for technical writers and travel agents; conversely, real estate managers will specialize in complex negotiations and stakeholder management, shrinking employment while increasing wages.⁵

However, our education-based measure differs from Autor and Thompson’s expertise concept: their framework would label some tasks as high expertise where ours specifies low education—for example, the Electrician task “Connect wires to circuit breakers, transformers, or other components.” And these predictions are based on current Claude usage patterns, which will shift as models are trained on new capabilities and users discover new applications—potentially changing which tasks are covered and whether the net effect is deskilling or upskilling.

Revisiting the aggregate productivity implications of Claude usage

In earlier work, we [estimated that widespread adoption of AI could increase US labor productivity growth by 1.8 percentage points](#) annually over the next decade. Here we revisit that analysis, incorporating the task success primitive introduced in this report and a richer treatment of task complementarity.

Based on the speedups associated with tasks with at least 200 observations in our sample of 1M Claude.ai conversations,⁶ we replicate our previous finding that current-generation AI models and current usage patterns imply a productivity effect of 1.8 percentage points per year over the next decade.⁷

With the inclusion of 1P API data, we can assess whether implied labor productivity effects differ based on enterprise Claude deployment patterns. Two countervailing forces are at play: API usage is more concentrated in a narrower set of tasks and occupations (particularly coding-related work), which would tend to reduce implied effects; but task-level speedups are higher on average among API tasks, as implied by Figure 4.1. These forces largely offset: the API sample likewise implies a 1.8 percentage point increase in labor productivity over the next decade.

A salient critique of this analysis is that it fails to account for model reliability. If workers must validate AI output, the productivity benefits will be smaller than raw speedups suggest. To assess how quantitatively important this channel might be, we incorporate the task success primitive introduced in this report, multiplying task-level time savings by task-specific success rates before aggregating.⁸

This adjustment has a meaningful effect: implied productivity growth falls from 1.8 to 1.2 percentage points per year for the next decade based on Claude.ai usage, and to 1.0 percentage points for API traffic. Yet, even after accounting for reliability, the implied impact remains economically significant—a sustained increase of 1.0 percentage point per year for the next ten years would return US productivity growth to rates that prevailed in the late 1990s and early 2000s.

A second critique concerns task complementarity. If some tasks are essential and cannot easily be substituted, then overall productivity effects will be constrained regardless of speedups on other tasks. Teachers may prepare